Ex No: 6

Develop a customized ping command to test server connectivity

Aim:

To develop a simple python program to check whether a server is reachable using the ping command.

Requirements:

Hardware:
* computer
* Internet connection

Software:
* python
* Vs code
* command prompt / Terminal

Theory:

-) ping is a network tool used to check whether a server is reachable

-) It uses ICMP to send requests to the destination and recive replies.

-) The customised ping program:

* Takes a website or IP address as input
* Sends 4 ping requests
* Displays ping result
* shows whether server is Reachable/ Unreachable.

program:

```
import subprocess
import platform
print("=" * 50)
print("Customized ping Utility")
print("=" * 40)
host = input("Enter website or IP Address: ")
if platform.system() == "Windows":
    command = ["ping", "-n", "4", host]
```

o/p:

```
Python ping.py
Enter website or IP address:
google.com
pinging google.com [142.250.193.78]
with 32 bytes
Reply from 142.250.193.78:
    bytes - 32 time = 28 ms TTL=116
Reply from 142.250.193.78:
    bytes = 32 time = 30 ms TTL=116
Reply from 142.250.193.78:
    bytes = 32 time = 27 ms TTL=116
Reply from 142.250.193.78:
    bytes = 32 time = 29 ms TTL=116
ping statistics for 142.250.193.78
packets:
    sent=4
    Received = 4
    Lost = 0 (0% Loss
Approximate round trip times
minimum = 27 ms
maximum = 30 ms
    Average = 28ms
Server Status: Reachable
```

```
else:
    command = ["ping", "-c", "4", host]
try:
    result = subprocess.run(
        command,
        capture_output = True,
        text=True)
    print("\n ping Results:\n")
    print(result.stdout)
    if result.returncode == 0:
        print("Server status:
                                Reachable")
    else:
        print("server status: unreachable\n")
except Exception as e:
    print("Error:", e)
```

Result:

Thus the customised ping command was developed in python to test server connectivity